

Supernode Analysis (Special Case in Nodal Analysis)

1. High-Level Overview & Big-Picture Intuition

What is a Supernode?

In standard Nodal Analysis, we write KCL equations by expressing branch currents in terms of node voltages using Ohm's Law ($I = \frac{\Delta V}{R}$). However, when an **ideal voltage source** (or a voltage source with no series resistance) is connected directly between two **non-reference nodes**, a dilemma arises:

- We cannot directly express the current flowing through an ideal voltage source using Ohm's Law, because an ideal voltage source has zero internal resistance ($R = 0$, leading to division by zero: $I = \frac{\Delta V}{0}$).

The Intuitive Solution: Enclose and Merge!

To bypass this problem, we group the two non-reference nodes together with the connected voltage source into a single generalized node called a **Supernode**.

- **Analogy:** Imagine two room doors connected by a fixed, rigid step-up platform (the voltage source). You don't know how many people walk back and forth inside the small hallway between the doors, but if you draw a big boundary surrounding both doors together, the total net number of people entering and leaving the combined two-room complex must still equal zero.

Key Rules of a Supernode:

1. **Combine Nodes:** Mentally encircle the voltage source and its two connecting non-reference nodes to treat them as a single **Supernode**.
2. **KCL Equation:** Write a single KCL equation for the entire Supernode boundary by summing all currents leaving all external branches connected to it.
3. **Constraint Equation:** Write a simple relationship equation provided directly by the voltage source between the two nodes:

$$V_{\text{positive terminal}} - V_{\text{negative terminal}} = V_{\text{source}}$$

2. Step-by-Step Method Breakdown

Step 1: Identify Voltage Sources Between Non-Reference Nodes

Look for any voltage source connected directly between two non-reference nodes (e.g., between V_1 and V_2).

Step 2: Form the Supernode

Draw a dashed boundary enclosing the voltage source and the two non-reference nodes attached to its terminals.

Step 3: Write the Constraint Equation

Relate the node voltages inside the Supernode using the voltage source value:

$$V_{\text{high}} - V_{\text{low}} = V_{\text{source}}$$

Step 4: Write the KCL Equation for the Supernode

Sum all currents leaving the combined Supernode boundary equal to zero:

$$\sum I_{\text{leaving supernode}} = 0$$

Step 5: Solve the System of Simultaneous Equations

Combine the Supernode KCL equation with the Constraint equation (and KCL equations from any remaining regular nodes) to determine all node voltages.

3. General Circuit Walkthrough & Derivation

Let us demonstrate how to construct the Supernode equations using the introductory example.

Diagram: Supernode Concept Circuit Layout

Visual Description:

The circuit consists of two essential non-reference nodes labeled V_1 and V_2 , connected by a horizontal 4 V independent voltage source (longer line/positive terminal on the left at V_1 , negative terminal on the right at V_2).

- Across the top, a $2\ \Omega$ resistor connects node V_1 to node V_2 .
- At node V_1 (left): a 2 A current source points upward into V_1 , and a $4\ \Omega$ resistor connects V_1 to Ground.
- At node V_2 (right): a 5 A current source points upward into V_2 , and an $8\ \Omega$ resistor connects V_2 to Ground.
- Bottom wire is connected directly to Ground (0 V).

Step 1: Constraint Equation

Looking at the 4 V source connected between V_1 and V_2 (positive terminal at V_1):

$$V_1 - V_2 = 4 \implies V_1 = V_2 + 4$$

Step 2: KCL at the Supernode (V_1, V_2)

Enclose V_1 , V_2 , and the 4 V source into a single Supernode boundary.

Sum all branch currents leaving this combined boundary:

1. **Current leaving V_1 via 2 A current source:** -2 A (since 2 A enters).
2. **Current leaving V_1 down through $4\ \Omega$ resistor:** $\frac{V_1 - 0}{4} = \frac{V_1}{4}$.
3. **Current leaving V_1 through top $2\ \Omega$ resistor towards V_2 :**

Note: Branches connected directly between the two nodes inside the supernode (like the top $2\ \Omega$ resistor) cancel out internally when summing currents leaving the combined boundary!

4. **Current leaving V_2 down through $8\ \Omega$ resistor:** $\frac{V_2 - 0}{8} = \frac{V_2}{8}$.
5. **Current leaving V_2 via 5 A current source:** -5 A (since 5 A enters).

Sum of all currents leaving the Supernode = 0:

$$\begin{aligned}\frac{V_1}{4} - 2 + \frac{V_2}{8} - 5 &= 0 \\ \frac{V_1}{4} + \frac{V_2}{8} &= 7\end{aligned}$$

Multiply entire equation by 8 to clear fractions:

$$2V_1 + V_2 = 56$$

Step 3: Solve for V_1 and V_2

Substitute $V_1 = V_2 + 4$ into $2V_1 + V_2 = 56$:

$$2(V_2 + 4) + V_2 = 56$$

$$2V_2 + 8 + V_2 = 56$$

$$3V_2 = 48 \implies V_2 = 16 \text{ V}$$

Now find V_1 :

$$V_1 = 16 + 4 = 20 \text{ V}$$

4. Comprehensive Worked Example

Problem Statement

In the circuit shown below, determine the current flowing through the 20Ω resistor.

 Diagram: Supernode Practice Problem Circuit

Visual Description:

Circuit components layout:

- **Bottom line:** Ground reference (0 V).
- **Left vertical branch:** Contains a 40Ω resistor at the bottom connected to Ground, leading up to a junction node. Above this junction is a 30 V voltage source (longer positive line on top). The top of the source connects via a horizontal 30Ω resistor to a top-center node.
- **Middle region:** A horizontal 10Ω resistor connects the left junction to a middle junction. Above the middle junction is a vertical 20Ω resistor going up to the top-center node. Below the middle junction is a vertical 10Ω resistor going down to Ground.
- **Right branch:** Top-center node connects to a vertical 20 V source (positive terminal on top) in series with a vertical 50Ω resistor going down to Ground.

Step-by-Step Solution

Step 1: Label the Nodes

Let us identify all primary junctions and label their node voltages:

- **Ground Node:** Bottom wire = 0 V .
- **Node 1 (V_1):** Junction between 40Ω resistor, 10Ω resistor, and bottom terminal of 30 V source.
- **Node 2 (V_2):** Junction above 30 V source (between 30 V source and 30Ω resistor).

Notice that V_2 and V_1 are connected by a 30 V source:

$$V_2 - V_1 = 30 \text{ V} \implies V_2 = V_1 + 30$$

(Nodes 1 and 2 form **Supernode 1-2**)

- **Node 3 (V_3):** Middle junction between 10Ω resistor, lower 10Ω resistor, and 20Ω resistor.
- **Node 4 (V_4):** Top-right junction connecting 30Ω resistor, 20Ω resistor, and upper terminal of 20 V source.
- **Node 5 (V_5):** Junction between negative terminal of 20 V source and top of 50Ω resistor.

Notice that V_4 and V_5 are connected by a 20 V source:

$$V_4 - V_5 = 20 \text{ V} \implies V_5 = V_4 - 20$$

(Nodes 4 and 5 form **Supernode 4-5**)

Step 2: Simplify Practical Branches / Write Node Equations

Let's evaluate Supernode 1-2, Node 3, and Supernode 4-5:

1. Supernode 1-2 (Nodes V_1 and V_2):

- Constraint equation:

$$V_2 = V_1 + 30$$

- KCL at Supernode 1-2 (sum of currents leaving):

$$\frac{V_1 - 0}{40} + \frac{V_1 - V_3}{10} + \frac{V_2 - V_4}{30} = 0$$

Substitute $V_2 = V_1 + 30$:

$$\frac{V_1}{40} + \frac{V_1 - V_3}{10} + \frac{V_1 + 30 - V_4}{30} = 0$$

Multiply through by 120 (LCM of 40, 10, 30):

$$3V_1 + 12(V_1 - V_3) + 4(V_1 + 30 - V_4) = 0$$

$$3V_1 + 12V_1 - 12V_3 + 4V_1 + 120 - 4V_4 = 0$$

$$19V_1 - 12V_3 - 4V_4 = -120 \quad \text{--- (Equation 1)}$$

2. Node 3 (V_3):

- KCL at Node 3 (sum of currents leaving):

$$\frac{V_3 - V_1}{10} + \frac{V_3 - 0}{10} + \frac{V_3 - V_4}{20} = 0$$

Multiply through by 20:

$$2(V_3 - V_1) + 2V_3 + (V_3 - V_4) = 0$$

$$-2V_1 + 5V_3 - V_4 = 0 \quad \text{--- (Equation 2)}$$

3. Supernode 4-5 (Nodes V_4 and V_5):

- Constraint equation:

$$V_4 - V_5 = 20 \implies V_5 = V_4 - 20$$

- KCL at Supernode 4-5 (sum of currents leaving):

$$\frac{V_4 - V_2}{30} + \frac{V_4 - V_3}{20} + \frac{V_5 - 0}{50} = 0$$

Substitute $V_2 = V_1 + 30$ and $V_5 = V_4 - 20$:

$$\frac{V_4 - (V_1 + 30)}{30} + \frac{V_4 - V_3}{20} + \frac{V_4 - 20}{50} = 0$$

Multiply through by 300 (LCM of 30, 20, 50):

$$10(V_4 - V_1 - 30) + 15(V_4 - V_3) + 6(V_4 - 20) = 0$$

$$-10V_1 - 300 + 10V_4 + 15V_4 - 15V_3 + 6V_4 - 120 = 0$$

$$-10V_1 - 15V_3 + 31V_4 = 420 \quad \text{--- (Equation 3)}$$

Step 3: Solve System of Equations

We now have 3 linear equations with 3 unknown voltages (V_1, V_3, V_4):

$$\begin{cases} 19V_1 - 12V_3 - 4V_4 = -120 & \text{--- (1)} \\ -2V_1 + 5V_3 - V_4 = 0 & \text{--- (2)} \\ -10V_1 - 15V_3 + 31V_4 = 420 & \text{--- (3)} \end{cases}$$

From Equation 2, express V_4 in terms of V_1 and V_3 :

$$V_4 = -2V_1 + 5V_3$$

Substitute V_4 into Equation 1:

$$19V_1 - 12V_3 - 4(-2V_1 + 5V_3) = -120$$

$$19V_1 - 12V_3 + 8V_1 - 20V_3 = -120$$

$$27V_1 - 32V_3 = -120 \quad \text{--- (Equation 4)}$$

Substitute V_4 into Equation 3:

$$-10V_1 - 15V_3 + 31(-2V_1 + 5V_3) = 420$$

$$-10V_1 - 15V_3 - 62V_1 + 155V_3 = 420$$

$$-72V_1 + 140V_3 = 420 \quad \text{--- (Equation 5)}$$

Solve Equations 4 and 5 simultaneously:

From Eq (4): $V_1 = \frac{32V_3 - 120}{27}$

Substitute into Eq (5):

$$-72 \left(\frac{32V_3 - 120}{27} \right) + 140V_3 = 420$$

$$-\frac{8}{3}(32V_3 - 120) + 140V_3 = 420$$

$$-\frac{256}{3}V_3 + 320 + 140V_3 = 420$$

$$\left(140 - \frac{256}{3}\right)V_3 = 100$$

$$\left(\frac{420 - 256}{3}\right)V_3 = 100 \implies \frac{164}{3}V_3 = 100$$

$$V_3 = \frac{300}{164} \approx 1.83 \text{ V}$$

Now find V_1 :

$$V_1 = \frac{32(1.83) - 120}{27} \approx -2.28 \text{ V}$$

Now find V_4 :

$$V_4 = -2(-2.28) + 5(1.83) = 4.56 + 9.15 = 13.71 \text{ V}$$

Step 4: Calculate Current Through 20 Ω Resistor

The 20 Ω resistor is connected between Node 4 ($V_4 = 13.71 \text{ V}$) and Node 3 ($V_3 = 1.83 \text{ V}$).

Using Ohm's Law:

$$I_{20\Omega} = \frac{V_4 - V_3}{20} = \frac{13.71 \text{ V} - 1.83 \text{ V}}{20 \Omega} = \frac{11.88 \text{ V}}{20 \Omega} = 0.594 \text{ A} = 594 \text{ mA}$$

5. Summary Cheat-Sheet

Method	When to Use	Key Formulation Steps
Standard Nodal Analysis	Independent/Dependent Current sources or Voltage sources with series resistors	1. Convert voltage sources to current sources. 2. Write KCL: $\sum \frac{\Delta V}{R} = 0$ at each node.
Supernode Analysis	Ideal Voltage source directly between two non-reference nodes	1. Group nodes & voltage source into a Supernode . 2. Write Constraint Equation : $V_{\text{high}} - V_{\text{low}} = V_{\text{source}}$. 3. Write 1 combined KCL equation for supernode boundary.